COMPUTER NETWORKS

UNIT 4 • NETWORKING DEVICES AND SERVERS

Complete, easy-language, exam-ready study notes

Prepared from the supplied Unit-4 lecture presentation. This guide follows its topics, explains the key ideas in plain English, and adds practice and revision aids.

Course: Computer Networks (CN) • DU #2304CS301

Unit outcome: Use networking devices and servers for establishing a computer network.

Contents

1. Unit overview and topic checklist

2. Networking devices

3. Types of servers

4. Exam preparation and question bank

5. Final revision and answer key

This document is organized by the source presentation’s outline: NIC, repeater, hub, modem and DSL/ADSL, switch, bridge, router, gateway, access point, then server types and six functional servers.

1. Unit overview and checklist

A **networking device** is hardware that connects computers and other network-enabled equipment so they can communicate and exchange data. Devices differ mainly in what they inspect (signals, MAC addresses, or IP addresses), where they work in the OSI model, and what action they take with traffic.

Source topic	Covered
Networking device fundamentals and OSI layers	✓
NIC / LAN card; MAC address; drivers; wired and wireless	✓
Repeater	✓
Hub, operation, characteristics and types	✓
Modem; location/media classifications; DSL and ADSL	✓
Switch, MAC learning/forwarding and types	✓
Bridge and bridge types	✓
Router, packet forwarding, features and route types	✓
Gateway	✓
Wireless access point and applications	✓
Server definition and dedicated/shared allocation	✓
File, application, print, mail, proxy and web servers	✓
Source question bank and added exam practice	✓

OSI model placement at a glance

The OSI model divides network communication into layers. The source explicitly places repeaters and hubs at Layer 1, traditional switches and bridges at Layer 2, routers at Layer 3, and protocol-converting gateways traditionally at the Application layer. A NIC participates mainly at Layers 1 and 2. A Layer 3 switch also routes at Layer 3. An access point bridges wireless clients to a LAN; its precise implementation may involve more than one layer.

Device	Main information/action	OSI layer in source
Repeater	Regenerates signal	Physical (L1)
Hub	Repeats incoming signal to all other ports	Physical (L1)
Bridge	Filters/forwards frames using MAC addresses	Data Link (L2)
Switch	Forwards frames using MAC table	Usually L2; L3 switch also routes
Router	Routes packets using IP/routing table	Network (L3)
Gateway	May translate protocols	Depends on function; traditionally application-level for protocol conversion

2. Networking devices

2.1 NIC / LAN card

Definition: A Network Interface Card (NIC), also called a LAN card or network adapter, is the hardware interface that lets a computer or other device join a network.

How it works: The NIC provides the wired Ethernet or wireless Wi-Fi connection. A device driver lets the operating system control the hardware; many modern systems detect and configure compatible NICs automatically through Plug and Play. Each interface has a MAC (Media Access Control) address, a physical identifier used for local-network delivery.

• A standard MAC address is 48 bits (6 bytes), written as six hexadecimal pairs, for example 06:02:03:04:4C:2B.

• A device may have separate wired and wireless network interfaces, each with its own MAC address.

• A NIC works mainly with physical transmission and data-link framing: the source presentation associates it with OSI Layers 1 and 2.

Remember / avoid confusion

A MAC address identifies a network interface on a local link; it is not the same as an IP address, which routers use to move packets between networks. Do not assume every network adapter is a separate expansion card: modern computers commonly build it into the motherboard or device.

2.2 Repeater

Definition: A repeater receives a weakened or degraded signal, regenerates it, and retransmits it so communication can reach farther.

Signals can lose strength or quality as they travel along a medium. A repeater restores the signal's usable form at the physical level and sends it onward. A typical repeater has two ports and connects two network segments; it is not a multi-device traffic selector.

• OSI Layer 1 (Physical).

• Extends communication distance by signal regeneration.

• Does not inspect MAC addresses or choose a destination device.

2.3 Hub

Definition: A hub (Ethernet hub, repeater hub, or concentrator) connects devices in a physical star and repeats an incoming signal out through every other port.

Example: If A sends data into a five-port hub with ports A–E, the hub sends it to B, C, D, and E. The intended computer accepts the data; the others see the signal but normally ignore data not meant for them. A hub does not learn or store MAC addresses, so it is called a non-intelligent device in the lecture.

• Works at Layer 1. Common port counts in the material are 2, 4, 8, or 16.

• Works half-duplex: a connected device cannot send and receive at the same time on the shared medium.

• All hub ports share one collision domain, so simultaneous transmissions can collide.

• Passive hub: provides physical connection and passes a signal without regenerating/amplifying it.

• Active hub: regenerates or amplifies the received signal before forwarding.

• Intelligent hub: adds monitoring and management features to connectivity.

Hub versus switch

Feature	Hub	Switch
OSI layer	Layer 1	Usually Layer 2; some also Layer 3
Decision basis	Does not inspect MAC addresses	Learns/uses MAC address table
Forwarding	Sends incoming signal to all other ports	Known unicast goes to destination port; unknown destination is flooded
Collision domains	All devices share one	Each port is a separate collision domain
Duplex in lecture	Half-duplex	Can support full-duplex in typical switched Ethernet

2.4 Modem

Definition: A modem—modulator-demodulator—is a signal-translating device that connects a home or office network to an Internet Service Provider (ISP) through an access medium.

Modulation converts data into a signal format suitable for transmission over the chosen medium; **demodulation** recovers data from the incoming signal. In traditional telephone-line examples, digital data is modulated into an analog signal and incoming analog signals are demodulated back into digital data. Modern cable, DSL, fiber, and cellular equipment implement the corresponding physical signaling for their media.

Modem classifications from the lecture

• By physical location: internal modem (inside a computer/device) and external modem (separate unit).

• By connection medium: cable, DSL, fiber-optic, satellite, wireless/cellular, and dial-up modems.

DSL and ADSL

DSL (Digital Subscriber Line) provides Internet access over traditional copper telephone lines. It can allow telephone service and Internet access at the same time. **ADSL** (Asymmetric DSL) is one member of the DSL family; its download rate is generally higher than its upload rate, making it suitable when users download more than they send.

Feature	DSL	ADSL
Meaning	Digital Subscriber Line: family of technologies	Asymmetric Digital Subscriber Line: a DSL type
Speed pattern	May be symmetric or asymmetric, depending on type	Download generally faster than upload
Use	General term for Internet over copper telephone lines	Useful when download demand exceeds upload demand

Exam caution: ADSL is not a technology separate from DSL; it is a specific kind of DSL. The source describes DSL as a family, so do not claim that all DSL types have identical upload/download speeds.

2.5 Switch

Definition: A switch connects devices such as computers, printers, and servers within a LAN and forwards frames toward the appropriate port. A traditional switch is Layer 2; a Layer 3 switch also performs routing functions.

MAC-learning and forwarding process:

A frame arrives. The switch reads its source MAC address and records that address with the incoming port in its MAC address table.

The switch looks up the destination MAC address.

If the destination is known, the switch sends the frame only through the mapped port.

If the destination is unknown, the switch floods it through active ports other than the incoming port.

The switch continues learning and updating its table as traffic arrives.

A switch supports unicast (one sender to one receiver), multicast (one sender to a selected group), and broadcast (one sender to all devices in the broadcast domain). The lecture illustrates a table mapping port numbers 1–4 to MAC addresses; that table is the switch's learned forwarding information.

Switch types

Classification	Type	Meaning
OSI layer	Layer 2	Uses MAC addresses to forward frames within a LAN.
OSI layer	Layer 3	Combines switching with IP routing at Layer 3.
Management	Unmanaged	Plug-and-play; little or no configuration; often used in small networks.
Management	Managed	Configurable features such as VLANs and monitoring; useful in larger networks.
Special feature	PoE switch	Carries data and supplies power over Ethernet to compatible equipment such as IP cameras and Wi-Fi access points.

Important distinction: A switch is sometimes called a multiport bridge because both can make Layer-2 forwarding decisions. Unlike a hub, it makes those decisions using MAC information.

2.6 Bridge

Definition: A bridge connects LAN segments at Layer 2. It examines frames and uses MAC addresses to decide whether to forward them to another segment or filter (discard) them, reducing unnecessary traffic between segments.

A two-port bridge separates the network into two collision domains, one on each side. It does not simply strengthen a signal: that is a repeater's job. A bridge makes a frame-forwarding decision.

Bridge type	Description
Transparent bridge	Common in Ethernet LANs; hosts generally do not need to know it is present. It learns MAC addresses and decides whether to forward or filter frames.
Source-routing bridge	Associated in the lecture with Token Ring. The sending computer specifies the path a frame should take through the bridge network.

2.7 Router

Definition: A router is an internetworking device that forwards packets between different networks. It operates at Layer 3 and uses destination IP addresses and a routing table to select where packets should go.

Packet-forwarding sequence: Receive a packet → inspect its destination IP address → consult the routing table → choose the best available route → forward the packet through the selected interface/next hop.

• Connects separate networks, commonly a LAN to an Internet/WAN connection.

• May assign/manage local IP addresses in common home-network setups, and forwards packets.

• Routing protocols (the source gives RIP, Routing Information Protocol, as an example) can help routers exchange route information.

• Static routes are manually configured; dynamic routing updates routes using routing protocols.

• The source notes router configurations with 2, 4, 8, or more ports and says routers are generally more expensive than hubs, bridges, and switches. Port counts and prices vary by product.

• The source lists Cisco, D-Link, TP-Link, and ASUS as example manufacturers.

Router versus switch

Feature	Switch	Router
Typical scope	Devices within a LAN	Different IP networks
Address used for forwarding	MAC address	IP address
Data unit in this unit	Frame	Packet
OSI layer	Usually Layer 2	Layer 3

A router may contain an integrated switch and wireless access point, but these are distinct networking functions. A Layer 3 switch also routes, usually within enterprise networks.

2.8 Gateway

Definition: A gateway connects networks and serves as an entry or exit point. When networks use different protocols or architectures, a gateway may translate between them so they can communicate.

A gateway's OSI layer depends on its function. The lecture notes that a protocol-converting gateway is traditionally associated with the Application layer. In everyday networking, "default gateway" often means the router/interface a host uses to reach other networks; context determines the intended meaning.

Distinguish: A router forwards packets between networks using IP routes; a protocol gateway can perform translation between unlike systems. The terms may overlap in practical equipment, but their functions are not identical.

2.9 Wireless access point

Definition: A Wireless Access Point (WAP) lets Wi-Fi devices connect to a network. It bridges wireless clients to the wired LAN or other network and provides a central connection point for a WLAN (Wireless Local Area Network).

• Common settings in the source: home networks, offices/businesses, public Wi-Fi hotspots, manufacturing/warehousing, smart homes and IoT, transportation hubs, and campuses.

• Large offices/buildings use access points to provide wireless coverage to many users or areas.

• An access point is not necessarily the same as a router: an AP provides Wi-Fi access; a router connects IP networks. A single product may combine both.

3. Types of servers

Definition: A server is a computer or system that provides resources, data, services, or applications to other computers (clients) over a network.

Allocation models

Model	Meaning	Typical consequence
Dedicated server	Resources such as CPU, RAM, storage, and network capacity are assigned to one organization, user, or application.	More exclusive control/resources, but the organization bears the allocated capacity.
Non-dedicated/shared server	Resources are shared among multiple users, organizations, or applications.	Resource use is shared; capacity and performance are divided among workloads.

The source also classifies servers by function: file, application, print, mail, proxy, and web servers. These roles can be provided by separate machines or combined on a system, depending on design.

3.1 File server

Definition: A file server is a computer or storage device dedicated mainly to storing and managing files for network users.

Users store, access, and retrieve shared files according to their permissions. Centralized storage makes it easier to manage and share organizational data. FTP (File Transfer Protocol) is one protocol that can transfer files between computers and a file server. A file server focuses on file storage and sharing rather than running application logic for every client.

3.2 Application server

Definition: An application server hosts, runs, and manages applications. It commonly sits between a client application and backend business logic or databases.

When a client sends a service request, the application server can execute application logic, access data, manage sessions, enforce security rules, and return a service response. The lecture gives Java, PHP, and Python application servers as examples of application-server environments/technologies.

File server vs application server: A file server primarily stores and shares files; an application server runs program logic and returns processed results.

3.3 Print server

Definition: A print server manages one or more printers and allows multiple network users to share them.

It receives print jobs, sends each job to the appropriate printer, and may organize pending jobs in a print queue. Printers may connect through USB, Ethernet, or wireless links. A queue helps sequence jobs when the printer is busy.

3.4 Mail server

Definition: A mail (email) server sends, receives, stores, and manages email over a network, often the Internet.

Protocol	Role in the lecture
SMTP (Simple Mail Transfer Protocol)	Sends/submits and transfers email from a client to a mail server and between mail servers.
POP3 (Post Office Protocol version 3)	Lets an email client retrieve messages from a mail server.
IMAP (Internet Message Access Protocol)	Lets an email client access/manage messages held on a mail server.

Memory aid: SMTP is for sending/transfer; POP3 and IMAP are for mailbox access/retrieval. A simplified path is sender/client → SMTP → sending mail server → SMTP → recipient mail server → POP3/IMAP → recipient's mail client.

3.5 Proxy server

Definition: A proxy server is an intermediary between a client and a target server. It receives a client request, forwards it to the target, and relays the response back.

The client communicates with the proxy rather than making a direct connection to the target in the described arrangement. Depending on the setup, one or both endpoints may not know a proxy is involved. The lecture focuses on this intermediary request/response role; it does not claim every proxy provides every possible security or privacy feature.

3.6 Web server

Definition: A web server is a computer/system (hardware and software) that stores website content and delivers it to users using HTTP or HTTPS.

Request/response flow: A user enters a URL (Uniform Resource Locator) in a browser → the browser sends an HTTP request → the web server responds with requested resources, such as HTML documents, images, CSS files, and other web content. Web servers are a foundation of the World Wide Web (WWW).

Server roles at a glance

Server	Main service	Helpful cue
File	Central file storage and sharing	Stores files
Application	Runs application logic and returns results	Processes requests
Print	Shares printers and manages print jobs/queues	Prints jobs
Mail	Sends, receives, stores and manages email	SMTP sends; POP3/IMAP access
Proxy	Relays client requests and server responses	Intermediary
Web	Delivers website resources via HTTP/HTTPS	Browser request, web response

4. Exam preparation

High-value points to remember

• Layer 1: repeater regenerates signals; hub repeats them to all other ports.

• Layer 2: bridge and traditional switch use MAC addresses to filter/forward frames. A switch learns source MAC-to-port mappings.

• Layer 3: router uses destination IP and a routing table to forward packets between networks.

• Gateway may translate protocols; its layer depends on the task.

• NIC connects a device to a network and has a MAC address; standard MAC length is 48 bits / 6 bytes.

• ADSL is a type of DSL and generally has faster download than upload.

• Hub has one shared collision domain; each switch port is a separate collision domain.

• SMTP sends/transfers email; POP3 and IMAP retrieve/access it.

• File server stores/shares files; application server executes logic; web server delivers web resources.

Commonly confused concepts

Concepts	Difference to state in an answer
Repeater vs bridge	Repeater regenerates signals at L1; bridge examines MAC-addressed frames at L2.
Hub vs switch	Hub repeats to all ports and does not learn MACs; switch learns MACs and selectively forwards known unicast frames.
Switch vs router	Switch typically forwards LAN frames using MAC addresses; router forwards packets between networks using IP addresses.
DSL vs ADSL	DSL is the family; ADSL is one asymmetric member.
Router vs gateway	Router selects IP routes; gateway may translate between unlike protocols. “Default gateway” can mean the host’s route out of its local network.
File vs application server	File server stores/shares files; application server runs program logic.
SMTP vs POP3/IMAP	SMTP sends/transfers email; POP3/IMAP access/retrieve mailbox email.
Access point vs router	AP connects Wi-Fi clients to a network; router connects networks. Consumer devices may combine both.

Likely exam questions

These are practice prompts based on the lecture’s question bank and listed topics, not predictions of what will appear on an exam.

• Write a short note on a NIC. Explain MAC address and the role of a driver.

• Define a repeater and compare it with a bridge.

• What is a hub? Explain its operation, characteristics, and types.

• Explain a modem and classify modem types. Compare DSL and ADSL.

• Explain how a switch learns MAC addresses and forwards frames. Describe switch types.

• Define a bridge and explain transparent and source-routing bridges.

• Explain router operation, features, and static versus dynamic routing.

• What is a gateway? How does it differ from a router?

• Write a short note on an access point and list its applications.

• Define a server. Compare dedicated and shared servers. Explain the six functional server types.

• Explain the role of SMTP, POP3, and IMAP in email.

• Describe how a browser and web server exchange web resources.

Short-answer practice with answers

Q. What does NIC stand for? **A.** Network Interface Card. It is the interface that connects a device to a network.

Q. How long is a standard MAC address? **A.** 48 bits, or 6 bytes, usually written as six hexadecimal pairs.

Q. What is a hub's forwarding behavior? **A.** It repeats an incoming signal to all other ports and does not select a destination by MAC address.

Q. What does an active hub do? **A.** It regenerates or amplifies the received signal before forwarding it.

Q. What is flooding on a switch? **A.** Forwarding a frame with an unknown destination MAC through all active ports except the incoming port.

Q. Which address does a router use to route packets? **A.** The destination IP address, checked against its routing table.

Q. What makes ADSL asymmetric? **A.** Its download speed is generally higher than its upload speed.

Q. Which protocol is mainly used to send email? **A.** SMTP.

Q. What is a print queue? **A.** A list of print jobs waiting to be processed by a printer.

Q. What is a proxy server? **A.** An intermediary that forwards client requests to a target server and relays responses.

5. Final revision

Chapter-wise rapid recap

Topic	One-line recall
NIC	Network interface; driver enables OS control; MAC identifies interface on local network.
Repeater	Regenerates and retransmits a weakened signal; L1.
Hub	L1 multiport repeater; sends to all other ports; shared collision domain.
Modem	Modulates/demodulates signals to communicate over an access medium.
DSL / ADSL	DSL family over copper telephone lines; ADSL is asymmetric, usually download-faster.
Switch	Learns MAC-to-port mappings and forwards frames; usually L2.
Bridge	Connects LAN segments and filters/forwards frames using MAC addresses; L2.
Router	Routes packets between networks using IP and routing table; L3.
Gateway	Entry/exit and, where needed, protocol translation between networks.
Access point	Provides Wi-Fi clients access to a LAN/WLAN.
File server	Central file storage and sharing.
Application server	Runs application logic and returns results.
Print server	Shares printers and queues jobs.
Mail server	Manages email; SMTP sends, POP3/IMAP access.
Proxy server	Intermediary that relays requests/responses.
Web server	Responds to HTTP/HTTPS requests with website resources.

Definitions and keywords

MAC address: A 48-bit (6-byte) link-layer identifier for a network interface. **Collision domain:** A part of a shared medium in which transmissions can collide. **Flooding:** Sending a frame through multiple ports when the switch does not know the destination port. **Modulation:** Converting data into a transmission-suitable signal form. **Demodulation:** Recovering data from a received signal. **Routing table:** Information used by a router to select a path/next hop. **WLAN:** Wireless Local Area Network. **Client/server:** A client requests a service; a server provides it.

Essential numerical facts and protocol names

• MAC address: 48 bits = 6 bytes; six hexadecimal pairs.

• Hub ports mentioned: commonly 2, 4, 8, or 16.

• Switch port examples in the lecture: 4, 8, 16, 24, 48, and others.

• SMTP: Simple Mail Transfer Protocol. POP3: Post Office Protocol version 3. IMAP: Internet Message Access Protocol.

• FTP: File Transfer Protocol. HTTP: Hypertext Transfer Protocol. HTTPS: secure HTTP. URL: Uniform Resource Locator. RIP: Routing Information Protocol.

Last-minute checklist

■ Can I identify each device's main function and OSI layer?

■ Can I distinguish signal regeneration, MAC-frame forwarding, and IP-packet routing?

■ Can I explain switch learning, known-destination forwarding, and unknown-destination flooding?

■ Can I compare DSL and ADSL accurately?

■ Can I identify the main role of every server type?

■ Can I explain the email protocol direction: SMTP sends, POP3/IMAP access?

■ Can I draw or describe the basic browser–web-server request/response sequence?

MCQ practice

1. Which device regenerates a weakened signal at OSI Layer 1?

A. Router B. Repeater C. Bridge D. Mail server

Answer: B. A repeater operates at the Physical layer and regenerates signals.

2. What does a traditional Layer 2 switch use to forward frames?

A. MAC address B. URL C. Email subject D. Telephone number

Answer: A. A Layer 2 switch uses destination MAC information and its MAC table.

3. What happens when a switch does not know the destination MAC port?

A. It deletes every frame B. It floods other active ports except the input port C. It changes the source IP D. It sends the frame to a mail server

Answer: B. Unknown unicast traffic is flooded so the destination can receive it.

4. Which statement about ADSL is correct?

A. It is not part of DSL B. Upload is always faster C. Download is generally faster than upload D. It uses only fiber

Answer: C. ADSL is an asymmetric DSL type, generally with higher download speed.

5. Which device routes packets between IP networks?

A. Hub B. Router C. Repeater D. Passive bridge

Answer: B. A router uses destination IP and routing information.

6. Which protocol is mainly used to send/transfer email?

A. SMTP B. POP3 C. IMAP D. FTP

Answer: A. SMTP handles sending/submission and transfer between mail servers.

7. Which server manages print jobs and queues?

A. Proxy server B. Print server C. Web server D. Application server

Answer: B. A print server shares printers and manages incoming jobs.

8. Which server primarily stores and shares files?

A. File server B. Mail server C. Router D. Access point

Answer: A. File servers provide centralized file storage and sharing.

9. A Wi-Fi access point mainly connects wireless clients to what?

A. A network/LAN B. A print queue only C. A routing protocol D. A MAC table only

Answer: A. An AP provides wireless access to a network.

10. Which server acts as an intermediary between a client and target server?

A. File B. Proxy C. Print D. Web

Answer: B. A proxy forwards requests and relays responses.

Long-answer structure

For a “Explain switch” answer: define a switch; state usual Layer 2 operation; explain source-MAC learning and MAC table; describe known unicast forwarding and unknown-destination flooding; mention unicast/multicast/broadcast; cover Layer 2/Layer 3, managed/unmanaged, and PoE types; conclude with its advantage over a hub.

For a “Explain router” answer: define router and Layer 3; describe packet arrival, destination-IP lookup, routing-table decision, and forwarding; explain connectivity, addressing, routing protocols, static/dynamic routes; distinguish it from a switch and gateway.

For “Explain server types”: define client/server; distinguish dedicated and shared allocation; explain each role in a compact table; state protocol examples for file transfer, email, and web access where relevant.

Scope and accuracy note: This guide is based on the supplied presentation. It preserves the lecture’s syllabus and clarifies wording where useful. Device capabilities and OSI-layer descriptions can vary across implementations; the exam-oriented layer labels above follow the presentation. Image placeholders in the source did not expose readable diagram details, so diagrams have been described in words rather than reconstructed.